



CS753 - Practical Training

DFARM: A Deadline-Aware Fault-Tolerant Scheduler for Edge Computing

Implementation in Edge Layer

By

Ganji Sridhar Babu

252CS012

M. Tech - 3rd Semester

Department of Computer Science and Engineering

National Institute of Technology Karnataka, Surathkal.

In Guidance with

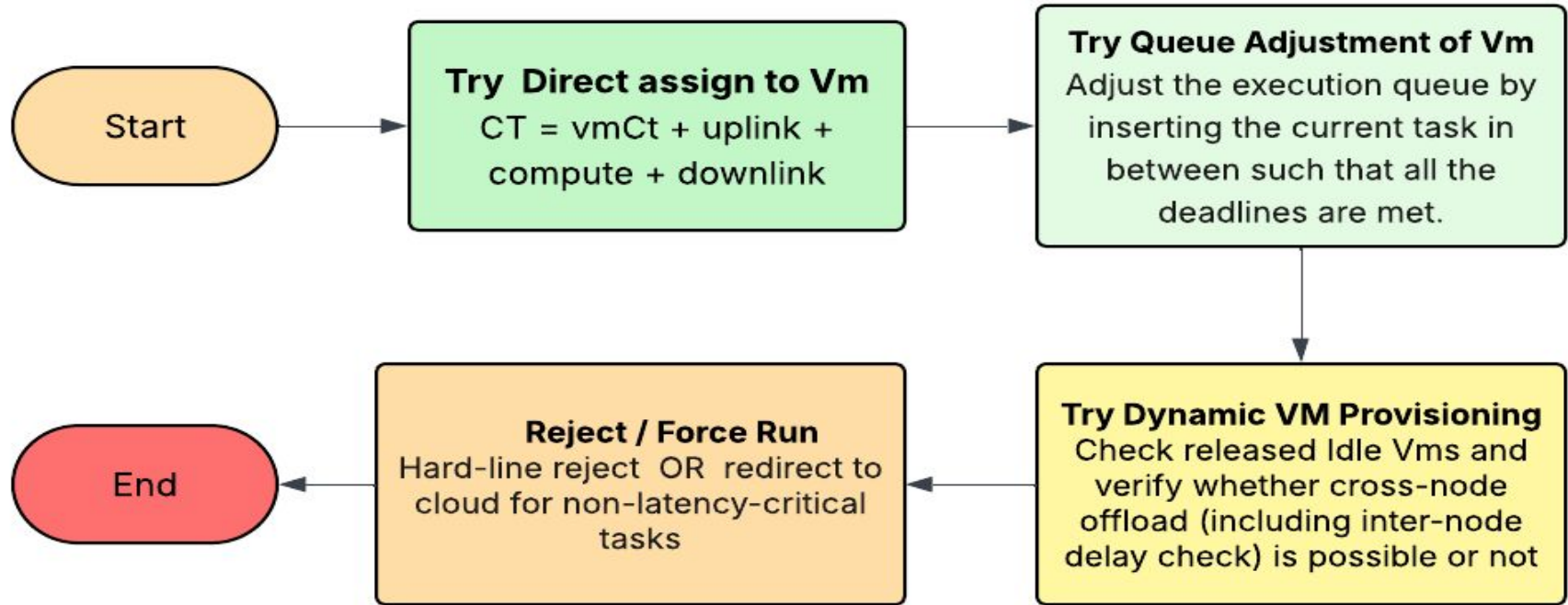
Prof. Annappa B

Department of Computer Science and Engineering

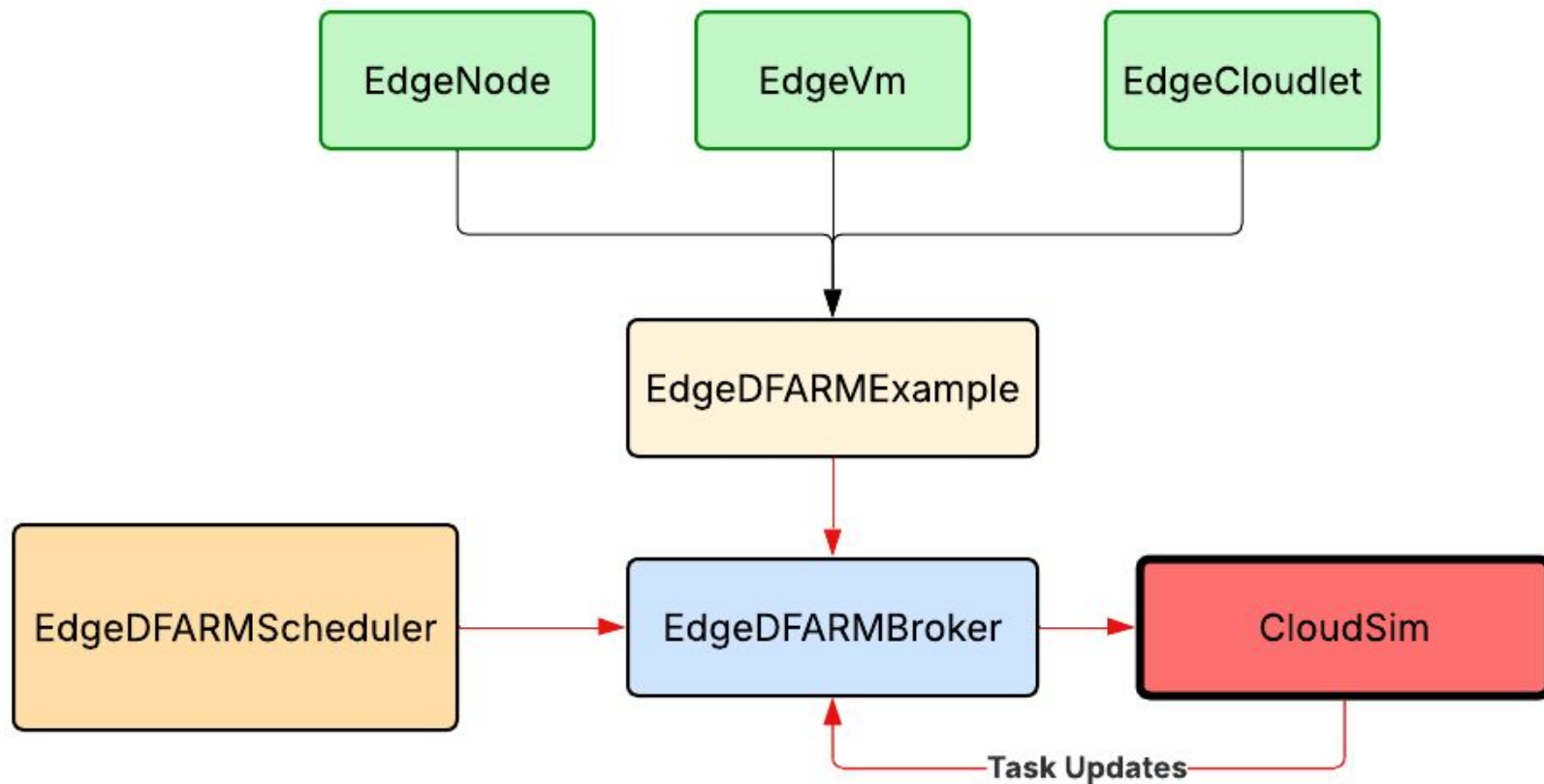
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DFARM: Architecture & Scheduling Logic

DFARM (Deadline and Fault-Aware Resource Manager) is an edge scheduler that allocates tasks to distributed edge nodes by jointly weighing task deadlines, cross-node offloading, replication, and network delays.



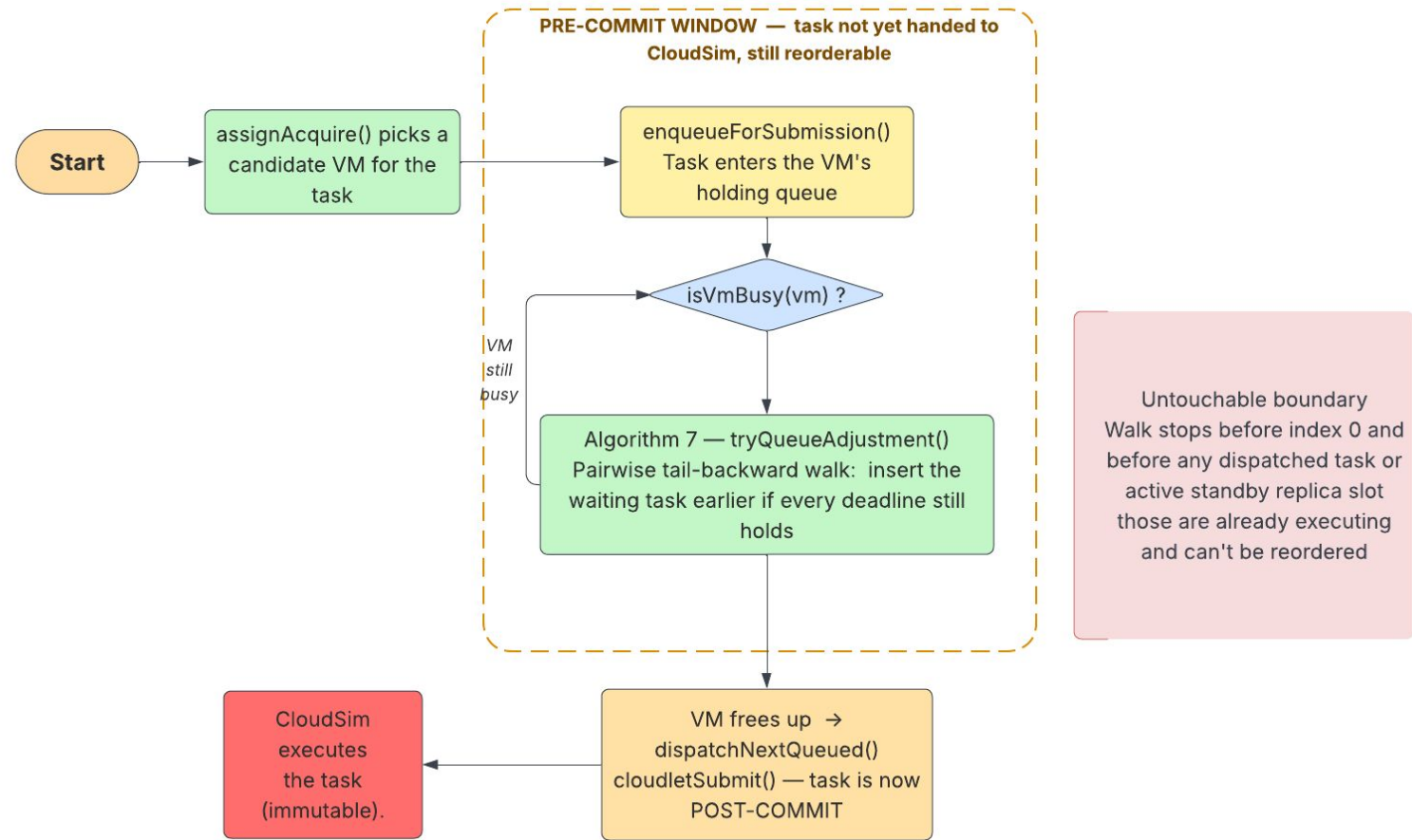
Design Overview



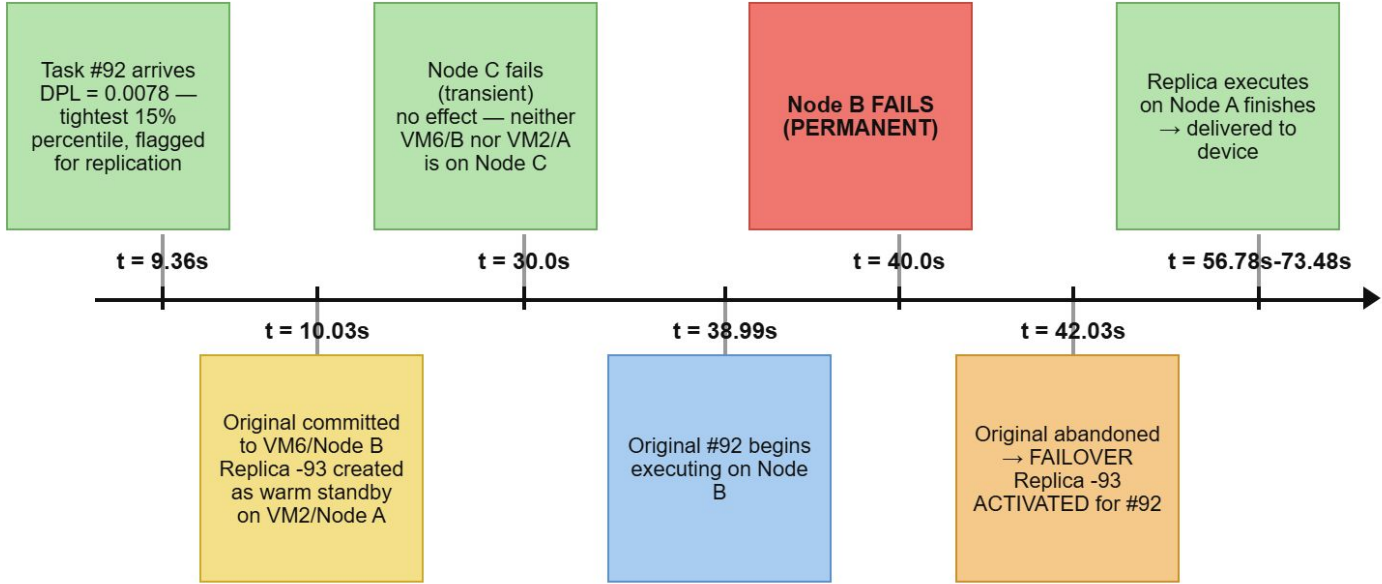
Enhancements over the Base CloudSim Implementation

- **Pre-Commit Phase for Queue Adjustment:** Added a per-VM holding queue that defers `cloudletSubmit()` until dispatch, opening a reorderable window where pairwise tail-backward walk can prioritize not-yet-submitted tasks to meet more deadlines, no such reordering existed in the base CloudSim scheduler.
- **Node Diversity:** Verified node-diversity replica placement end-to-end. every standby replica is guaranteed to sit on a different physical node from its original, confirmed across all 11 replica pairs in testing (100% cross-node diversity).
- **Node Failure Simulation:** Built a fault-injection framework (transient and permanent node failures with configurable timing) plus full failover instrumentation, capturing both a genuine replica-triggered recovery (task #92, 27.27% replica utilization) and the honest baseline case where replicas don't overlap a failure.
- **Time Complexity Analysis:** Instrumented the scheduler's VM-search path with real wall-clock timing (`System.nanoTime()`), characterizing per-task cost as $O(V \cdot \alpha + Q)$ and empirically confirming overhead stays under 0.05% of a task's deadline budget even in the worst-case cascade fallback.

Pre-Commit Phase: Enabling Queue Reordering



Replica Failover



First genuine failover: task #92 recovered via replica -93, ~31s after Node B went down | Run total: 3/11 replicas activated (27.27% utilization) — up from 0% when Node B failed at t=80s

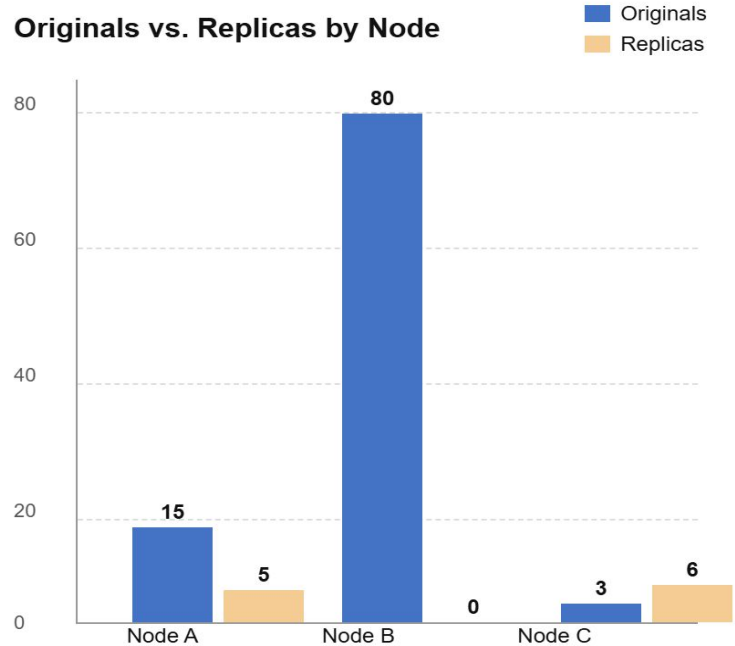
Scheduling Complexity & Timing Overhead

Per- task scheduling cost: Direct Assign $O(V_{node}) \rightarrow$ Queue Adjustment/Algorithm 7 $O(Q) \rightarrow$ Dynamic VM Provisioning $O(N \cdot V)$: overall worst case $O(V_{total} + Q)$ per task , linear in VM count and local queue depth, never combinatorial. Measured wall-clock search times (below) confirm this stays sub-millisecond on the direct-assign path and under ~30 ms even on the worst-case cascade fallback.

Task Tier	Task Count	Avg Deadline	Avg Search Time (ms)	Min Search Time (ms)	Max Search Time (ms)	Deadline Used (avg case)	Deadline Used (worst case)
Small (≤ 1000 MI)	12	34.76 s	0.991	0.043	3.368	0.0029%	0.0097%
Medium (2000 MI)	27	54.54 s	0.329	0.021	4.893	0.0006%	0.0090%
Large (≥ 5000 MI)	64	83.67 s	0.634	0.013	30.445	0.0008%	0.0364%
Overall	103	70.33 s	0.596	0.013	30.445	0.0008%	0.0433%

Node Diversity: Originals vs. Replica Placement

Originals vs. Replicas by Node



All 11 Standby-Replica Pairs (Fault-Injection Run)

Original Task	Original Node	Replica Cloudlet	Replica Node
#4	B	-5	C
#26	B	-27	A
#44	B	-45	C
#78	B	-79	A
#80	B	-81	A
#92	B	-93	A
#144	B	-145	A
#206	B	-207	C
#214	B	-215	C
#232	B	-233	C
#244	B	-245	C

11 of 11 replicas (100%) placed on a different physical node from their original

Experimental Setup

Simulator: iFogSim

Programming Language: Java

Cloudlet Types:

Category	Million Instructions
Small	1000MI
Medium	2000MI
Large	5000MI
Huge	8000MI

Edge Nodes and VMs:

Node Name	No of VMs	MIPS	Boot time`	Acquisition Delay
Node A	4	100,200,300,500	5s	10s
Node B	6	500,750, 900, 1000,1100, 1200	3s	8s
Node C	4	100,250,500,750	8s	15s

CloudSim : Terminal

DFARM iFogSim Edge Simulation Results

```
=====
Total tasks (original) : 250
Accepted                : 95
Rejected                : 155
Structural replica fails: 0
TRR (Task Rejection)   : 0.6200
Makespan                : 134.31 s
Throughput              : 0.7073 tasks/s
ARUR                   : 0.3473
=====
```

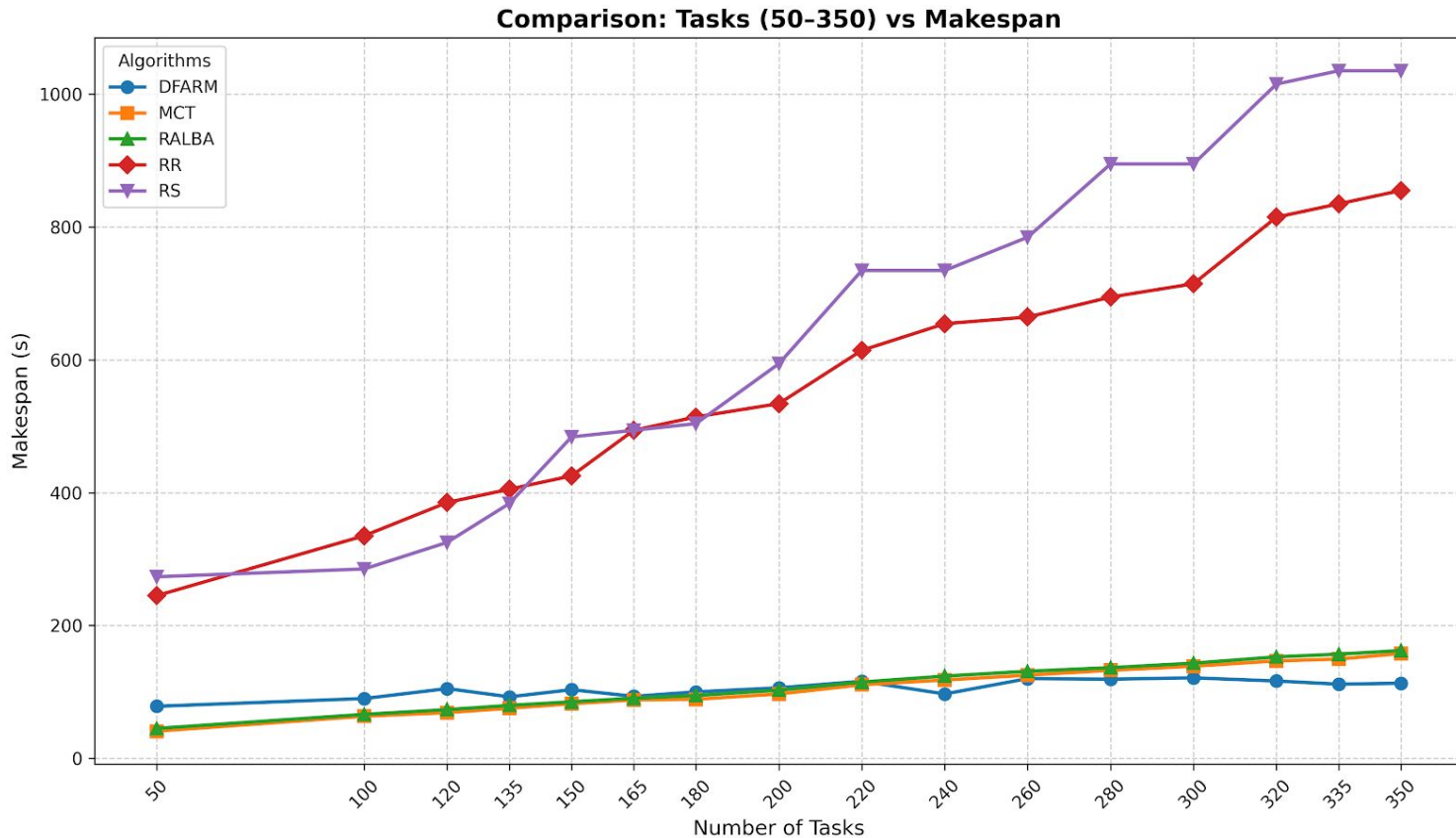
Fault tolerance / failure injection:

```
=====
Node failures injected : 2
Successful failovers   : 3
Reactive resubmissions : 3
Tasks lost             : 34
Standby replicas created: 11 (activated=3, discarded=3, lost-on-standby=5)
Replica utilization    : 27.27%
Extra cost of replication: 134.28 VM-seconds reserved
Avg recovery latency   : 2.0300 s
Avg replica activation : 30.03 s (standby idle time before use)
Availability (failed nodes): 57.4455%
MTTR                   : 20.00 s
=====
```

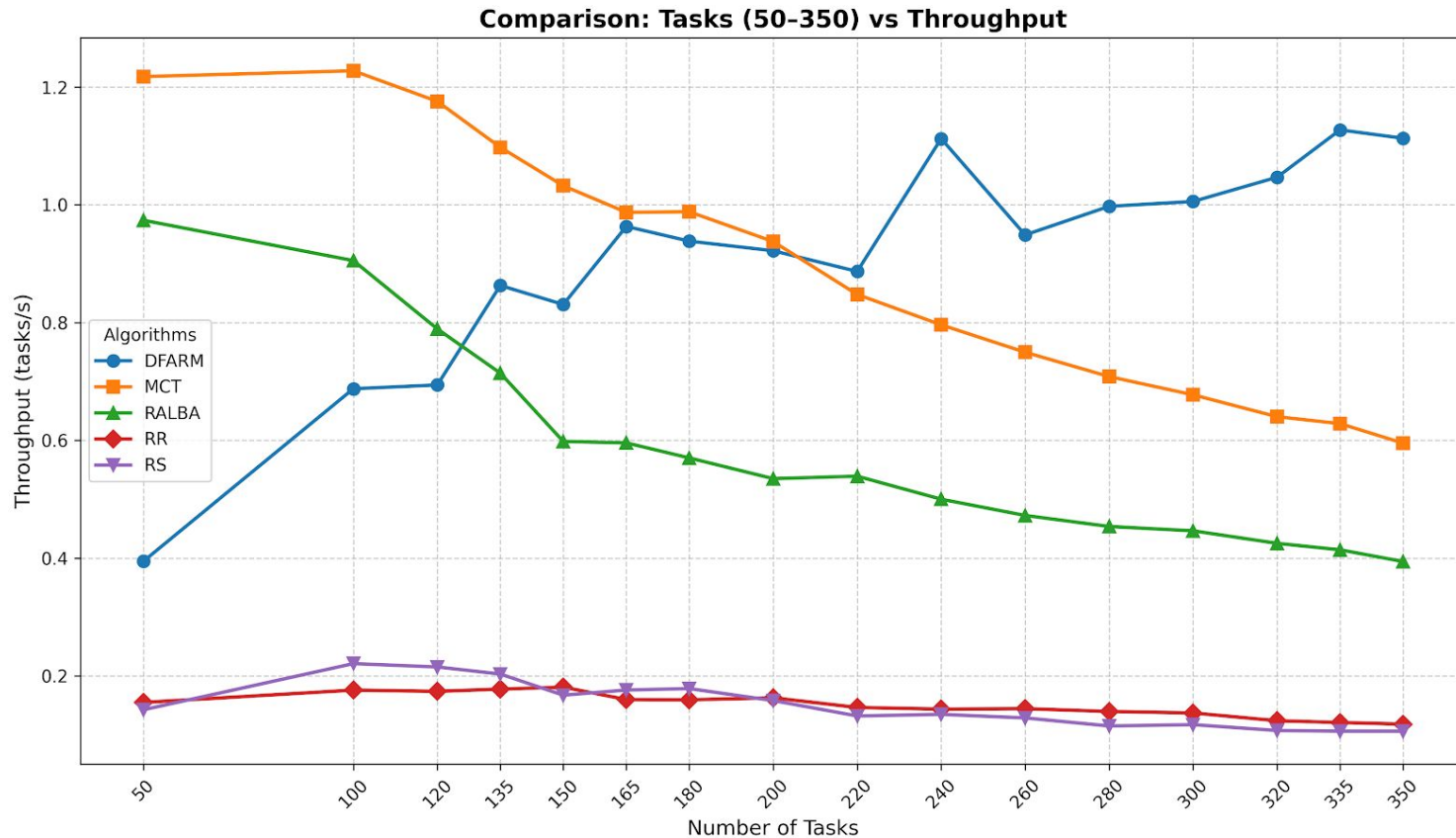
[Verification] Failover behavior check:

```
Node failures injected : 2
Successful failovers   : 3
Reactive resubmissions : 3
Tasks lost             : 34
Avg recovery latency   : 2.0300s
MTTR                   : 20.0000s
Lost task IDs: 37, 211, 195, 181, 214, 145, 219, 31, 23, 200, 59, 9, 147, 65, 49, 83, 1, 189, 64, 205, 244, 27, 47, 81, 121, 191, 193, 85, 89, 229, 44, 225, 171, 237
Node C (TRANSIENT) ΓÇö recovery reintegration exercised, not checked for zero residual (legitimate new work can land there again after recovery).
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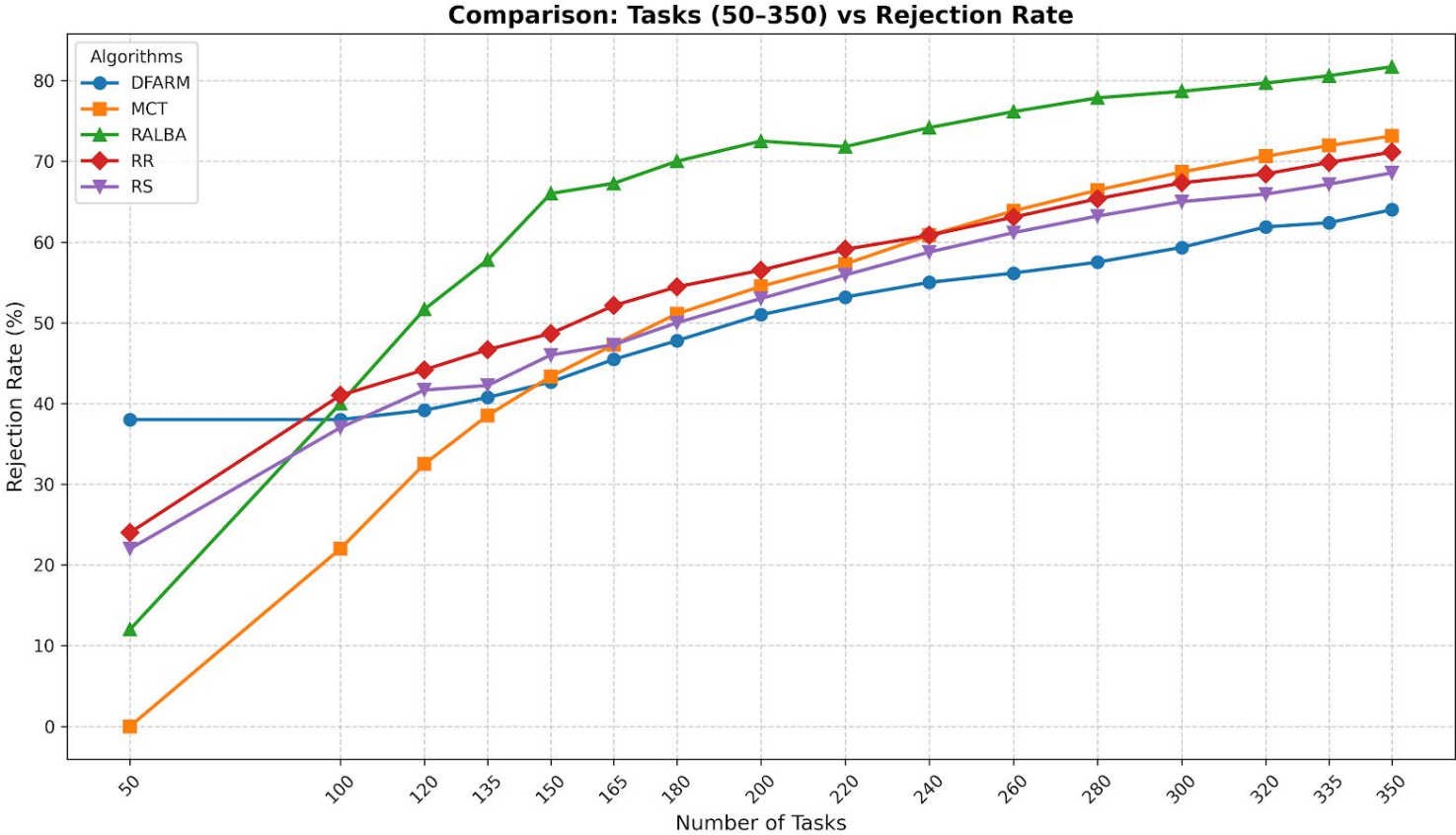
DFARM Results : MakeSpan



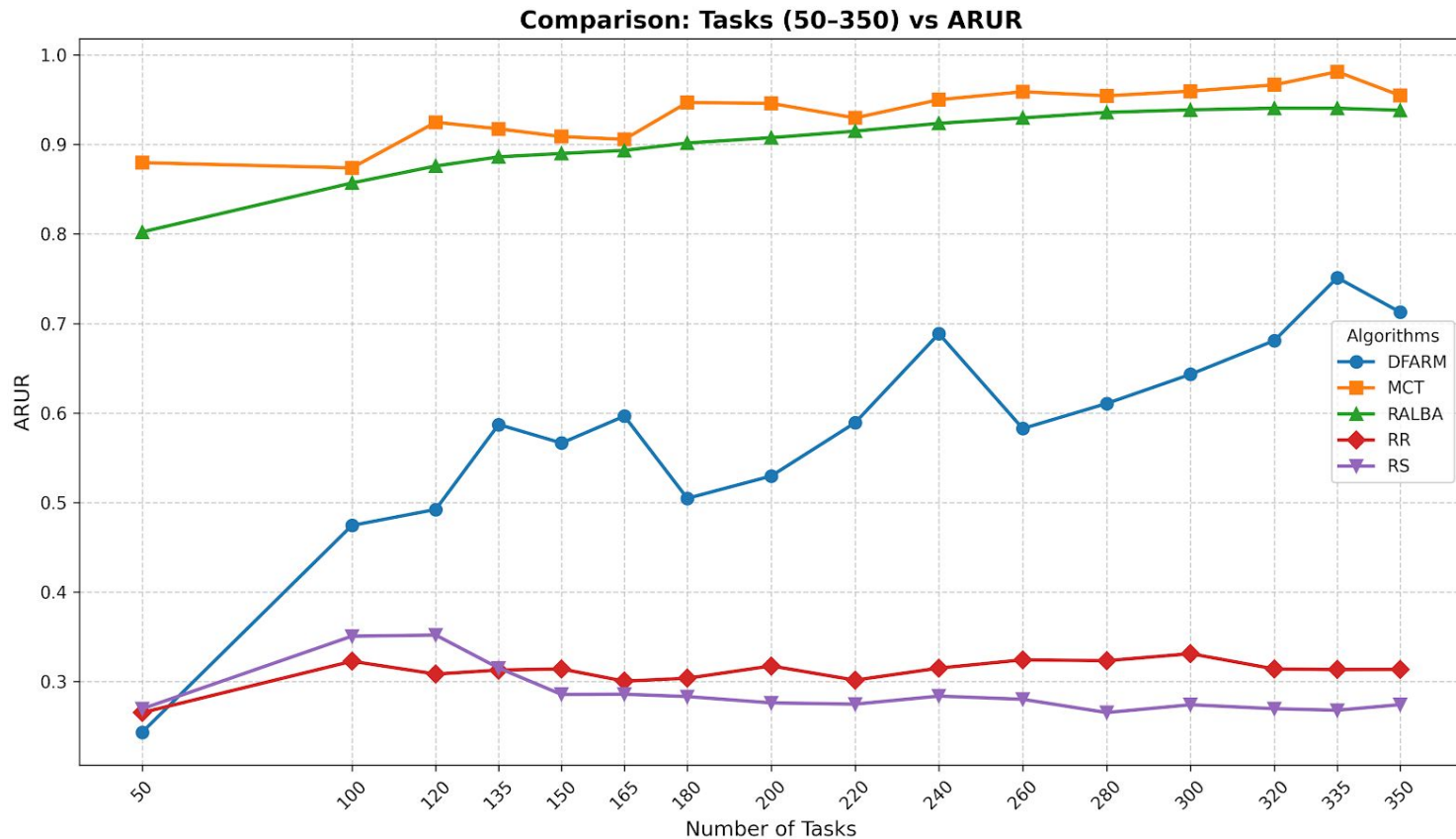
DFARM Results : Throughput



DFARM Results : Rejection Rate(%)



DFARM Results : Average Resource Utilization Ratio



References

1. Ahmad Awan, Muhammad Aleem, Altaf Hussain, Radu Prodan. "**DFARM: a deadline-aware fault-tolerant scheduler for cloud computing**." Cluster Computing, vol. 27, pp. 9323–9344, 2024. DOI: 10.1007/s10586-024-04419-1
1. Rodrigo N. Calheiros, Rajiv Ranjan, Anton Beloglazov, César A. F. De Rose, Rajkumar Buyya. "**CloudSim: A Toolkit for Modeling and Simulation of Cloud Computing Environments and Evaluation of Resource Provisioning Algorithms**." Software: Practice and Experience, vol. 41, no. 1, pp. 23–50, 2011. DOI: 10.1002/spe.995
1. Altaf Hussain, Muhammad Aleem, Arshad Khan, Muhammad Iqbal, Ansar Islam. "**RALBA: a computation-aware load balancing scheduler for cloud computing**." Cluster Computing, vol. 21, pp. 1667–1680, 2018. DOI: 10.1007/s10586-018-2856-x
1. Tracy D. Braun, Howard Jay Siegel, Noah Beck, Ladislau L. Bölöni, Muthucumaru Maheswaran, Albert I. Reuther, James P. Robertson, Mitchell D. Theys, Bin Yao, Debra Hensgen, Richard F. Freund. "**A Comparison of Eleven Static Heuristics for Mapping a Class of Independent Tasks onto Heterogeneous Distributed Computing Systems**." Journal of Parallel and Distributed Computing, vol. 61, no. 6, pp. 810–837, 2001. DOI: 10.1006/jpdc.2000.1714
1. Akash Garg, Ritu Garg. "**Fault Tolerance Mechanisms in Cloud Computing: A Survey**." International Journal of Computer Applications, vol. 143, no. 11, pp. 14–18, 2016. DOI: 10.5120/ijca2016910239

References

6. Tracy D. Braun, Howard Jay Siegel, Noah Beck, Ladislau L. Bölöni, Muthucumaru Maheswaran, Albert I. Reuther, James P. Robertson, Mitchell D. Theys, Bin Yao, Debra Hensgen, Richard F. Freund. "**A Comparison of Eleven Static Heuristics for Mapping a Class of Independent Tasks onto Heterogeneous Distributed Computing Systems.**" *Journal of Parallel and Distributed Computing*, vol. 61, no. 6, pp. 810–837, 2001. DOI: 10.1006/jpdc.2000.1714

THANK YOU
